Kubernetes, an open-source container orchestration platform, uses a range of parameters and options to manage clusters, nodes, pods, and other resources. Below is a comprehensive list of commonly used Kubernetes parameters, along with their definitions and descriptions:

General Parameters

`kubectl` Command Options

- **`-n, --namespace`**
  - **Definition**: Specifies the namespace scope for the command.
  - **Description**: Limits the scope of the command to the specified namespace.

- **`-o, --output`**
  - **Definition**: Output format (e.g., json, yaml, wide).
  - **Description**: Specifies the format in which to display output.

- **`-f, --filename`**
  - **Definition**: Filename, directory, or URL to files identifying the resource.
  - **Description**: Points to the file or directory containing resource definitions.

- **`--context`**
  - **Definition**: The name of the kubeconfig context to use.
  - **Description**: Switches between different clusters or contexts defined in the kubeconfig file.

- **`--kubeconfig`**
  - **Definition**: Path to the kubeconfig file to use for CLI requests.
  - **Description**: Uses a specific kubeconfig file for cluster configuration.

- **`--dry-run`**
  - **Definition**: If true, only print the object that would be sent, without sending it.

  - **Description**: Simulates the command without making any changes.

- **`--force`**
  - **Definition**: Force a command to execute without confirmation.
  - **Description**: Overrides safety checks to force the command.

Resource Definitions

Pod Parameters

- **`apiVersion`**
  - **Definition**: Defines the versioned schema of the object.
  - **Description**: Specifies the API version for the resource.

- **`kind`**
  - **Definition**: The type of Kubernetes resource.
  - **Description**: Defines the resource type (e.g., Pod, Service, Deployment).

- **`metadata`**
  - **Definition**: Metadata for the resource.
  - **Description**: Contains information such as name, namespace, labels, and annotations.

- **`spec`**
  - **Definition**: Specification of the desired behavior.
  - **Description**: Defines the desired state of the resource.

- **`containers`**
  - **Definition**: List of containers within a Pod.
  - **Description**: Specifies the containers that will run in the Pod.

- **`image`**
  - **Definition**: Docker image to run.
  - **Description**: Specifies the container image to use.

- **`ports`**
  - **Definition**: List of ports to expose.
  - **Description**: Defines the ports on which the container will listen.

- **`resources`**
  - **Definition**: Resource requests and limits.
  - **Description**: Specifies CPU and memory requirements.

- **`env`**
  - **Definition**: List of environment variables.
  - **Description**: Sets environment variables for the container.

- **`volumeMounts`**
  - **Definition**: List of volumes to mount.
  - **Description**: Defines the volumes and mount paths within the container.

Service Parameters

- **`spec.selector`**
  - **Definition**: Selector for choosing which pods the service will target.
  - **Description**: Matches labels to determine the set of pods the service routes traffic to.

- **`spec.ports`**
  - **Definition**: List of ports that the service will expose.
  - **Description**: Defines the service's port configuration.

- **`type`**
  - **Definition**: Type of the service (e.g., ClusterIP, NodePort, LoadBalancer).
  - **Description**: Determines how the service is exposed.

Deployment Parameters

- **`replicas`**
  - **Definition**: Number of desired pods.
  - **Description**: Specifies the number of pod replicas.

- **`strategy`**
  - **Definition**: Deployment strategy.
  - **Description**: Defines the strategy for updating pods (e.g., RollingUpdate, Recreate).

- **`template`**
  - **Definition**: Pod template.
  - **Description**: Specifies the pod configuration for the deployment.

ConfigMap Parameters

- **`data`**
  - **Definition**: Key-value pairs.
  - **Description**: Contains the configuration data.

Secret Parameters

- **`type`**
  - **Definition**: Secret type.
  - **Description**: Specifies the type of secret (e.g., Opaque, kubernetes.io/tls).

- **`data`**
  - **Definition**: Secret data.
  - **Description**: Contains base64-encoded secret data.

Volume Parameters

- **`name`**
  - **Definition**: Volume name.
  - **Description**: Identifies the volume.

- **`hostPath`**
  - **Definition**: Host directory.
  - **Description**: Mounts a host directory into the pod.

- **`emptyDir`**
  - **Definition**: Empty directory volume.
  - **Description**: Creates a temporary directory for the pod.

- **`configMap`**
  - **Definition**: ConfigMap volume.
  - **Description**: Mounts a ConfigMap as a volume.

StatefulSet Parameters

- **`serviceName`**
  - **Definition**: Headless service managing the StatefulSet.
  - **Description**: Links the StatefulSet to a service.

- **`volumeClaimTemplates`**
  - **Definition**: Volume claim templates.

- **Description**: Defines persistent volume claims for the StatefulSet.

PersistentVolumeClaim Parameters

- **`accessModes`**
  - **Definition**: Access modes for the volume.
  - **Description**: Specifies how the volume can be mounted (e.g., ReadWriteOnce, ReadOnlyMany).

- **`resources.requests.storage`**
  - **Definition**: Storage request size.
  - **Description**: Specifies the size of the requested storage.

HorizontalPodAutoscaler Parameters

- **`minReplicas`**
  - **Definition**: Minimum number of replicas.
  - **Description**: Specifies the minimum number of pod replicas.

- **`maxReplicas`**
  - **Definition**: Maximum number of replicas.
  - **Description**: Limits the maximum number of pod replicas.

- **`metrics`**
  - **Definition**: Metrics for scaling.
  - **Description**: Defines the metrics used to scale the deployment.

Ingress Parameters

- **`spec.rules`**
  - **Definition**: List of host rules.

  - **Description**: Defines the routing rules for the ingress.

- **`spec.tls`**
  - **Definition**: TLS configuration.
  - **Description**: Specifies the TLS settings for secure connections.

NetworkPolicy Parameters

- **`podSelector`**
  - **Definition**: Selector for pods.
  - **Description**: Matches labels to select the pods to which the policy applies.

- **`policyTypes`**
  - **Definition**: List of policy types.
  - **Description**: Specifies the types of policies (e.g., Ingress, Egress).

- **`ingress`**
  - **Definition**: Ingress rules.
  - **Description**: Defines the inbound traffic rules.

- **`egress`**
  - **Definition**: Egress rules.
  - **Description**: Defines the outbound traffic rules.

This list covers many of the essential Kubernetes parameters. Kubernetes is highly configurable, and specific resources and controllers may have additional parameters and options. For the most comprehensive and up-to-date information, always refer to the official [Kubernetes documentation](https://kubernetes.io/docs/home/).